

Signatures of Comparison

Languages employ various strategies to express comparison which differ in their morphosyntactic and semantic properties (Beck et al., 2009). We experimentally test four syntactic strategies of comparison in English exemplified in (1) across four comparison contexts, and a range of relative gradable adjectives. The resulting unique signatures for each expression of comparison captures their context dependence and can be adapted to other strategies in English, and to strategies in other spoken and signed languages.

- (1) a. Emani is taller than Theodore. [Explicit]
 b. Emani is tall compared to Theodore. [Implicit]
 c. Theodore is tall, Emani is very tall. [Intensification]
 d. Theodore is tall  Emani is TALL  [Gesture]

Some languages have dedicated morphosyntax for comparison that can be used across a broad range of semantic contexts like the English *X-er*(1a). It simply establishes the ordering $x \succ y$ and as such is notably felicitous in a *crisp judgment context*, i.e. where there is a very small but non-zero difference in the degrees along the dimension in which the objects are compared. In contrast, the interpretation of other *implicit* comparison strategies can vary in terms of their sensitivity to contextual factors (Kennedy, 2007). A *compared to* construction (1b) is more sensitive to contextual factors, and the difference between x and y must exceed some contextually determined standard (Castillo-Gamboa et al., 2021), making them infelicitous in *crisp judgment contexts*. Similarly, *intensification* strategies like (1c) can support crisp judgments, but require both entities to be in the positive extension of the gradable predicate. We also test a novel comparative involving iconic co-speech gestures (1d) to evaluate the effect on context on the interpretation of co-speech gesture.

The following image shows the speeds of different animals.
 Answer the following questions based on your observations.



The cheetah is fast compared to the antelope.

- ☐ Agree
☐ Disagree

The turtle is slow compared to the antelope.

- ☐ Agree
☐ Disagree

Fig. 1: Sample of implicit strategy with *fast*

Experiment design. We tested signatures of four strategies of comparison (1) by recruiting native English-speaking participants (N = 120, via Prolific) to judge a given comparison strategy (tested between subjects) across six different adjectival scales (*tall*, *heavy*, *thick*, *fast*, *hot*, *old*). We crossed scales and comparison contexts in a Latin Square design (24 total trials per participant). Comparison contexts were constructed as follows: each adjectival scale included six values <1....6>, all clearly ordered and labeled with respect to the relevant dimension, presented in random order in a single image. We tested a crisp judgment context where the difference between the two objects being compared is very small but the positive value holds for both (e.g. <5,6>). We further tested contexts with a significant difference between two values on a shared direction of the scale in both the positive and

negative direction (e.g. both short <1,3>, both tall <4,6>). Finally we tested comparison across the median, i.e. between an object on the high and on the low end of the scale (e.g. <2,5>) where the simple positive cannot be entailed. We collected binary judgments (*Agree/Disagree*) fully counterbalancing lexicalization on each end of the scale (e.g. *short* in <1,3> and *tall* in <4,6>) and expected outcome based on order being consistent with the given context (agree/disagree, i.e. disagree that 4 is faster than 6) [See Fig 1].

Results: We fit our data with a mixed effects logistic model in R ($judgment \sim context * expectedoutcome * strategy + (1 | participant) + (1 | adjective)$), and found a main effect of comparative strategy with each exhibiting a different signature of effects (Fig 2). The *Explicit* strategy only had a main effect of expected outcome (Intercept $\beta = 4.401, p < 0.05$); *Implicit* had a main effect of expected outcome and an interaction between outcome and (crisp judgment) $<5,6>$ context ($\beta = -4.49, p < 0.005$); *Intensification* had a main effect on expected outcome ($\beta = -2.80, p < 0.005$), and interaction between outcome and (across the median) $<2,5>$ context ($\beta = -4.37, p < 0.005$). Finally, *Gesture* also had a main effect on expected outcome ($\beta = -3.46, p < 0.05$), as well as an interaction between outcome and context (across the median) ($\beta = -3.55, p < 0.005$). There were no other significant effects, including no effect of adjectival scale.

Discussion. We identify empirical signatures of four semantically distinct comparative strategies across four comparison contexts. **Explicit (-er):** as predicted, sensitive across the board to any asymmetric ordering of two objects with respect to some property and insensitive to distance along the scale or the contextual standard. **Implicit (compared to):** Shows decreased acceptability for the crisp judgment case $<5,6>$, supporting an analysis in which they are licensed only when the difference exceeds a contextually determined standard (Castillo-Gamboa et al., 2021). Neither strategy entails that the positive form is true, making them felicitous in the $<2,5>$ condition. **Intensification:** Comparatives of the form ‘A is tall, B is very tall’ assert the positive form of the gradable adjective to be true of both entities, and are predictably rejected in the $<2,5>$ (comparison across the median). **Gesture:** Similarly, gesture comparatives also entail the positive form to be true of both entities and are unacceptable in the $<2,5>$ condition. However, in the $<5,6>$ crisp judgment condition, participants were not sensitive to the difference in values expressed via co-speech gesture, judging both *Agree* and *Disagree* cases positively. But, we also see greater sensitivity to comparison via gesture as the distance between values increased in $<1,3>$ and $<4,6>$ conditions.

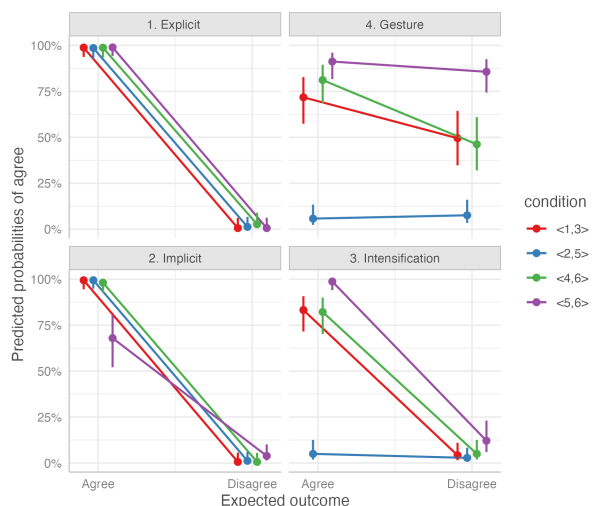


Fig. 2: Judgment by strategy

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Going forward. This experimental design motivates an approach to considering *families of strategies* within a given language to understand how their distribution may be constrained by context, particularly in languages that might lack an explicit strategy like *-er*. For example, a “degree-less” spoken language like Washo employs *šemu* (‘really’) strategy in the crisp judgment condition (Bochnak, 2015), but would differ from explicit comparatives in the across the median condition. In the domain of sign languages, this approach would allow us to evaluate the semantic import of iconic content in these constructions. With such a paradigm we can arbitrate between differing theories of iconicity such as those assuming an “iconic” degree (Aristodemo and Geraci, 2018) as opposed to a co-gestural approach (Thalluri and Davidson, 2024), with each predicting a different semantic signature for the iconic strategy of comparison.

References

- Aristodemo, V. and C. Geraci (2018). “Visible degrees in Italian sign language”. *Natural Language & Linguistic Theory* 36.3
- Beck, S. et al. (2009). “Crosslinguistic variation in comparison constructions”. *Linguistic variation yearbook* 9.1
- Bochnak, M. R. (2015). “The degree semantics parameter and cross-linguistic variation”. *Semantics and pragmatics* 8
- Castillo-Gamboa, J., A. Wellwood, and D. Rudin (2021). “Being tall compared to compared to being tall and being taller”. *Experiments in Linguistic Meaning* 1
- Kennedy, C. (2007). “Modes of comparison”. *Proceedings from the annual meeting of the Chicago Linguistic Society*. Vol. 43. 1. Chicago Linguistic Society
- Thalluri, N. and K. Davidson (2024). “Degrees and depiction—gradability in sign languages”. *Proceedings of Sinn und Bedeutung*. Vol. 28